

Invoice Pipeline — Detailed System Design v8

Companion to Design Reference v8. Power Automate orchestration; Logic Apps removed.

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1. System overview — components and zones

Three platform zones. Microsoft 365 holds the document landing zone and the human-facing queue and alerts. Azure holds the decision engine and AI extraction. Power Platform holds the orchestration, the review tooling, and the Dynamics 365 CE destination — the orchestrator is a Power Automate cloud flow (Logic Apps is not used). There is no Blob staging area (binary transport) and no duplicate-invoice check (requirement).

#	Flow	Detail
1	SharePoint trigger	Power Automate cloud flow polling trigger fires within ~1–5 min of upload
2	Fetch and call	Power Automate reads file content and the item UniqueId, then calls the Function through a custom connector
3	Decide	Function: gate A1 atomic claim, one binary call to the CU router (classify, route, extract), then gates
4	Log and return	Function writes Received then Extracted with the decision, and returns the decision
5	Write	Power Automate 'Add a new row' on the happy path; ledger finalized after the 201
6	Review item	Any failed gate: Power Automate writes an item to the SharePoint review queue instead

2. Component inventory

Component	Zone	Responsibility
SharePoint library	Microsoft 365	Invoice landing zone; one invoice per PDF; source of truth for files
Review queue list	Microsoft 365	InvoiceReviewQueue SharePoint list; flagged items with a status workflow
Microsoft Teams	Microsoft 365	Adaptive-card alerts to reviewers
Power Automate (orchestration flow)	Power Platform	Trigger, file fetch, Function call (custom connector), branch, Dataverse write, terminal ledger update
Azure Function	Azure	Decision engine: A1 atomic claim, binary CU call, gates B2–B4 and GST, ledger writes
Content Understanding	Azure	Router analyzer plus general invoice (and deferred property tax) child analyzers
Table Storage	Azure	InvoiceExtractProcessLog: gate A1 idempotency claim plus file-to-Dynamics audit ledger
App Insights	Azure	Function run history, latency, error telemetry
Entra ID	Azure	Function user-assigned managed identity; custom-connector federated credential for production
Power Automate (review flows)	Power Platform	Notify (item created) and approval (status = Approved) flows, off the review list
Power Apps	Power Platform	Review app: PDF beside extracted fields, low-confidence highlighting
Dynamics 365 CE	Power Platform	Dataverse invoice table; the final write destination

Authentication model

Three boundaries, authenticated independently. Switching the orchestrator to Power Automate changes the first two; the Function's own identity is unchanged.

Boundary	Mechanism	Notes
Power Automate → Function	Custom connector	Function key held in the connection for the prototype (encrypted at rest, not in run history); Entra federated-credential managed identity for production. The HTTP action has no managed identity, so a custom connector is used.
Power Automate → Dataverse / SharePoint	Connection references	A user or service-principal connection, not a managed identity. Credential ownership and rotation belong to the connection owner.
Function → CU / Table Storage	User-assigned managed identity	Unchanged. Cognitive Services User on the Foundry resource; Storage Table Data Contributor on the storage account.

Secret handling. Holding the function key in the custom connector's connection keeps it out of the flow definition and out of run history, and for a single key is usually simpler than a Key Vault round-trip (the Key Vault connector itself needs a service-principal or user connection). Move to the secretless custom-connector managed-identity path for production.

Routing gate reference

Gate	Stage	Condition	Action
A1	Pre (Function)	Item id already claimed and decided in the ledger	Skip — do not reprocess (idempotency)
B2	Post	Category = other	Reject
B3	Post	is_handwritten = yes (general invoice)	Always review
B4	Post	Critical field confidence below 0.75	Review
GST	Post	total minus GST not reconciling to 5%	Review
B6	Post	line_items row confidence below 0.65 (advisory)	Flag row; continue
B7	Post	All conditions clear	Power Automate adds the row to Dataverse

B1 is not present (no per-category confidence at GA); B5 is removed (no duplicate-invoice verification). B4 absorbs misclassification through depressed field confidence.

3. Human review subsystem

Power Automate runs the whole pipeline, so review is additional flows on the same platform as the spine, not a separate one. Both review flows trigger off the SharePoint review list: the notify flow posts the Teams card on item creation; the approval flow fires on status = Approved and re-enters the orchestration flow's Dataverse write action with the reviewer's corrected fields.

Single write owner. The approval path re-enters the orchestration flow's write action rather than writing to Dataverse independently. One owner for the Dataverse write keeps the ledger-after-row-ID rule from drifting. Both the happy path and the approval path converge on this same write action.

Function is extraction-only. On approval the pipeline does not re-invoke the Function or Content Understanding — extraction already happened and the reviewer corrected the fields, so the approval flow goes straight to the write action.

No spine/review isolation. Because both the spine and the review flows run on Power Automate, a Power Automate service issue affects processing and review alike — the isolation the Logic-Apps spine provided is gone. Plan alerting and a manual fallback for sustained outages.

Reject path. The reviewer sets status = Rejected; a trivial flow alerts whoever handles manual disposition. No pipeline involvement.

4. InvoiceExtractProcessLog

Keyed on the SharePoint item id. Gate A1 uses an atomic insert-claim so one upload is processed once. Written by two owners with a clean handoff: the Function writes Received then Extracted with the routing decision; the Power Automate flow writes the terminal Written plus the Dynamics row id, and only after a confirmed 201. Multiple writers are acceptable because the table is a processing ledger, not the system of record; the single-write-owner rule governs the Dataverse write.

Property	Type	Notes
PartitionKey	String	First characters of the sanitized item id (load spread)
RowKey	String	Sanitized SharePoint item UniqueId — the A1 idempotency key
Status	String	Received / Extracted / InReview / Written / Rejected
RoutingDecision	String	Gate outcome from the Function
DocumentType / IsHandwritten	String	Classification and modality — audit trail
DynamicsRecordId	String	Populated only after a confirmed write (design's DataverseRowId)
IngestedUtc / LastUpdatedUtc	DateTime	First sighting and last update
SourceFileUrl	String	SharePoint URL of the original PDF
ExpiresAt	DateTime	Drives the nightly purge

Retention and purge

Parameter	Recommendation	Rationale
Minimum retention	13 months	One financial year plus a year-end buffer
Preferred retention	Match AP audit policy (often 7 years)	The ledger is part of the processing audit trail
Purge cadence	Nightly Azure Function timer	Small batch deletes; Table Storage has no native TTL
Never purge	Rows with an empty DynamicsRecordId	Stuck-invoice records — investigate before deleting

5. Ingestion idempotency

Earlier designs used three layers to prevent duplicate Dynamics rows (A1 item id, B5 Dataverse query, and a Dataverse alternate key). With duplicate-invoice verification removed by requirement, only one layer remains, and its job is narrower: ingestion idempotency, not deduplication.

Layer	Mechanism	What it provides
A1 — atomic claim	Table Storage insert-claim on the SharePoint item id, pre-extraction	One upload is processed once, even under concurrent trigger replays; a stale Received row is resumed after the lease

Out of scope. Invoice-level duplicate detection is no longer performed. The same invoice arriving as a new SharePoint file (a new item id) or entered manually into Dynamics is not detected and creates a new Dynamics row. The Dataverse duplicate query and the alternate key are removed, and the CE-administrator alternate-key action is withdrawn.

Dataverse load. Removing the duplicate query drops the per-invoice Dataverse calls from about two to one (the write). Service-protection limits (~6,000 requests per user per 5 minutes) are comfortable at this volume; the connector auto-retries on 429.

6. End-to-end sequence flow

Reproduced from the Design Reference for completeness. The Function owns the path up to the decision; the Power Automate flow owns the trigger, the branch, and all writes.

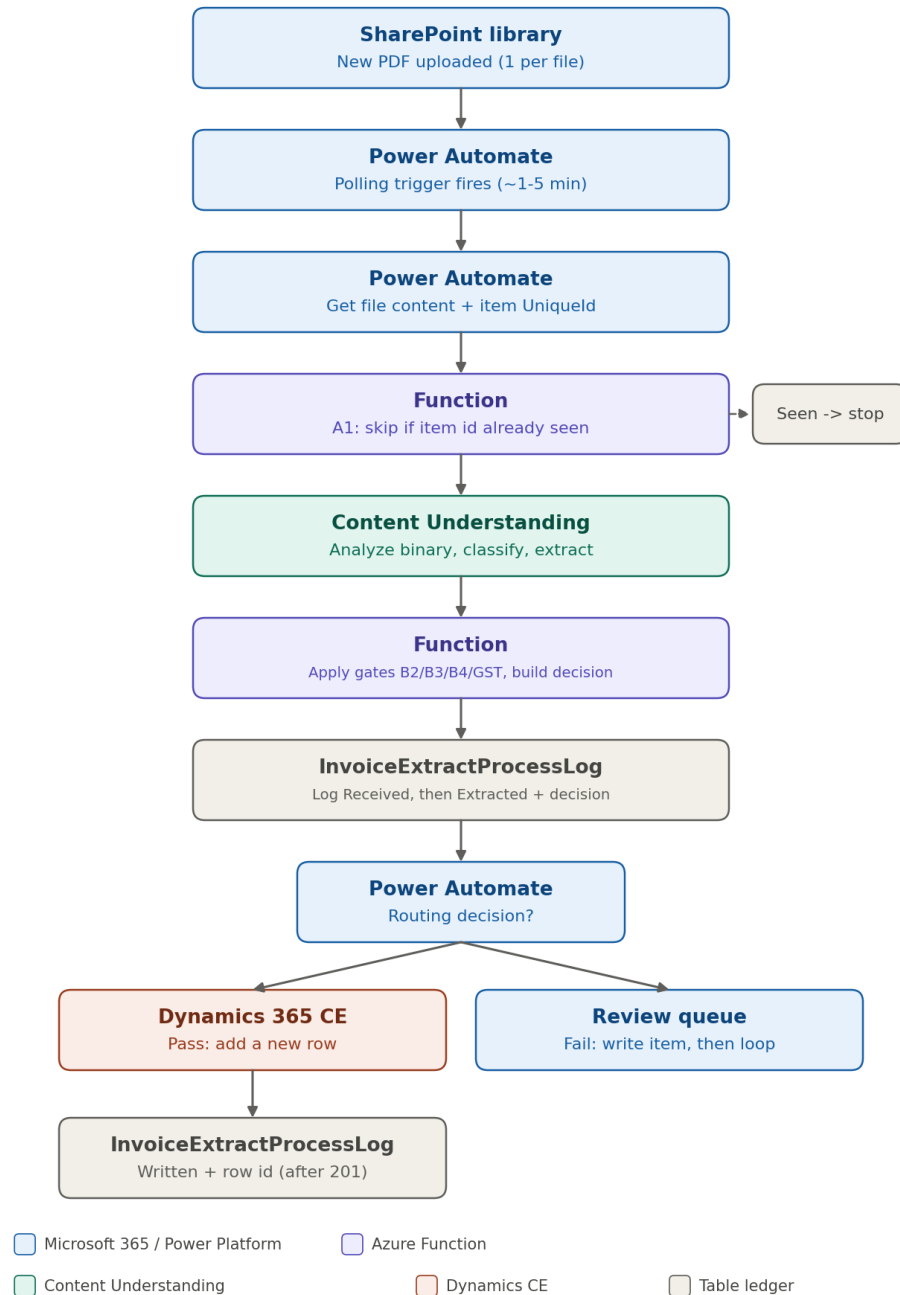


Figure 1. End-to-end sequence on the binary transport. Power Automate is the orchestrator — it calls the Function through a custom connector and writes to Dynamics through a Dataverse connection. Gate A1 is an atomic idempotency claim; the happy path writes directly. The ledger is written by the Function (Received, Extracted) and finalized by the Power Automate flow (Written plus row id) after the 201.